Assessment Tool - 2

Co-2

Scenario - Based Questions

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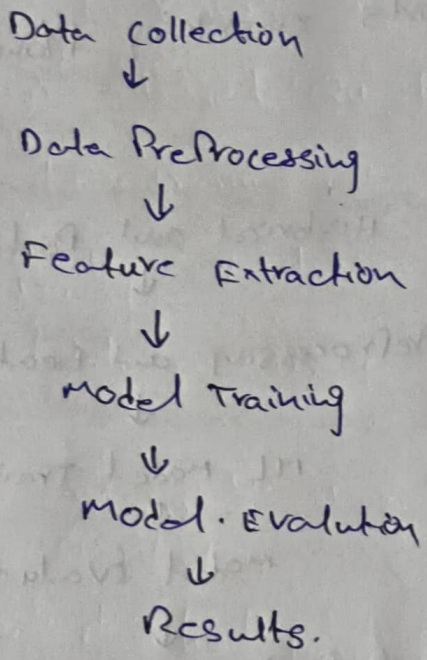
Course Code :- ITA0612

Course Name :- Machine Learning.

1) Explain the functional blocks of a Machine Learning System.

A machine learning (ML) system learns Patterns from data and makes Predictions.

Block diagram:-



Functional Blocks:-

1. Data collection:- Collect data from databases, sensors, and files.

2. Data PreProcessing:- Remove errors and handle missing values.

3. Feature Extraction:- Select important features from the data.

4. Model training:- Train the algorithm to learn Patterns.

5- Model Evaluation:- Check the model's Performance.

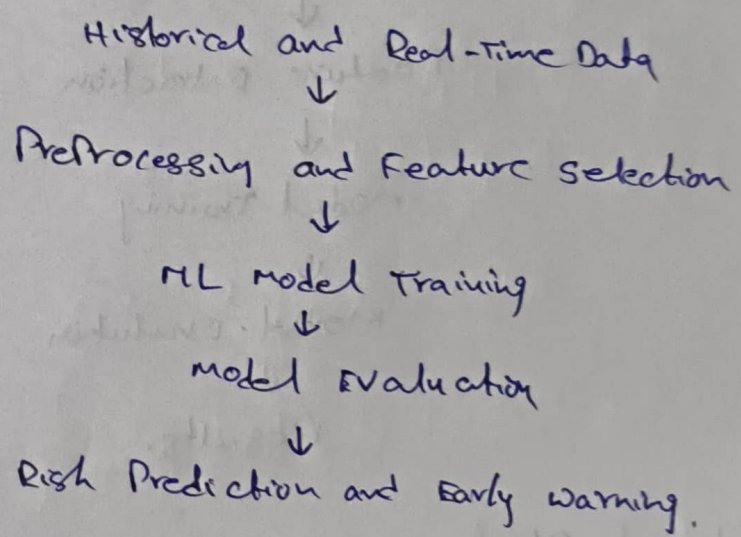
Conclusion:- These blocks help the ML system learn and Produce accurate Predictions.

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2) Design a machine learning system for disaster management.

A ML System helps Predict disaster risks such as floods and supports emergency Planning.

Block diagram:-



Steps:-

1) Data Collection:- Collect rainfall, river levels, weather and seismic data.

2) Preprocessing:- Remove duplicate records and handle missing values.

3) Feature selection:- Select important features such as rainfall intensity and river level.

4. Model selection:- Use neural networks to learn complex patterns & for simpler predictions.

5. Training and Evaluation:- Train using historical data and evaluate using accuracy, Precision and recall.

Applications:-

→ Provides early warnings
→ Identifies high risk areas.
→ Helps Plan evacuations and emergency responses.

Conclusion:-

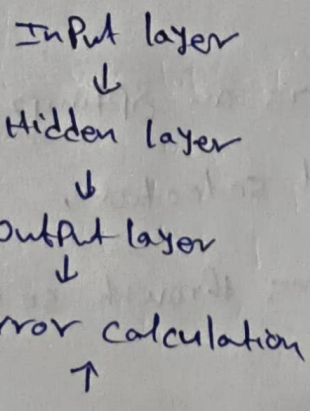
ML supports disaster management by improving risk assessment and preparedness.

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3) Explain the BackPropagation Algorithm in an MLP.

BackPropagation is a learning algorithm that reduces Prediction errors by updating neural network weights.

Block diagram:-



steps:-

1. Forward Pass:- Input data passes through the network to produce an output.

2. Error Calculation:- Compare the Predicted output with the actual output.

3. Backward Propagation:- Calculate error gradients from the output layer backwards.

4. weight Update:- Adjust weights using gradient descent

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$$

5. Repeat:- Continue until the error becomes sufficiently small.

Conclusion:- Back Propagation improves model accuracy through repeated weight updates.

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4) Explain the Genetic Algorithm using the given dataset.

A Genetic Algorithm is an optimization technique inspired by natural selection.

It improves solutions through selection, crossover and mutation.

Given dataset:-

Speed	Accuracy	Fitness	suitable
High	Low	40	No
Medium	medium	60	Yes
High	medium	75	Yes
Low	High	65	Yes
Medium	High	80	Yes

Steps:-

1. Initialization:- Start with the five candidate solutions.
2. Selection:- Choose candidates with higher fitness values, such as Example 3 and 5.
3. Crossover:- Combine characteristics of selected candidates to Produce new solutions.
4. Mutation :- Randomly change a characteristic to explore new solutions.
5. Evaluation:- Calculate the fitness of new candidates and repeat the Process.

Fitness improvement:-

candidate	Fitness
1	40
2	60
3	75
4	
5	80

Best solution:- Example 5 has Medium Speed High Accuracy and a fitness value of 80.

conclusion:- GA improve candidate solutions over generations.

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5) Explain neural network classification and evaluate its Performance:-

Architecture:-

Input:- Age, Blood Pressure, Glucose, lifestyle
↓
Hidden layer
↓
Output:- Disease Yes/No

Training steps:-

1) Preprocessing.
2) Forward Propagation
3) Error calculation.
4) Back Propagation.
5) Evaluation.

Confusion matrix values from the given data set:-
TP = 2 ; TN = 1 ; FP = 1 ; FN = 1.

Performance matrics:-

→ Accuracy = (TP+TN)/5 • x100 = 60%.
→ Precision = TP/(TP+FP) = 66.67%.
→ Recall = TP/(TP+FN) = 66.67%.
→ F1-score = 2PR/(P+R) = 66.67%.

Conclusion:- The model achieves 60% accuracy on the given five records.